Complex Molecular Assemblies at Hand via Interactive Simulations

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Abstract: Studying complex molecular assemblies interactively is becoming an increasingly appealing approach to molecular modeling. Here we focus on interactive molecular dynamics (IMD) as a textbook example for interactive simulation methods. Such simulations can be useful in exploring and generating hypotheses about the structural and mechanical aspects of biomolecular interactions. For the first time, we carry out low-resolution coarse-grain IMD simulations. Such simplified modeling methods currently appear to be more suitable for interactive experiments and represent a well-balanced compromise between an important gain in computational speed versus a moderate loss in modeling accuracy compared to higher resolution all-atom simulations. This is particularly useful for initial exploration and hypothesis development for rare molecular interaction events. We evaluate which applications are currently feasible using molecular assemblies from 1900 to over 300,000 particles. Three biochemical systems are discussed: the guanylate kinase (GK) enzyme, the outer membrane protease T and the soluble *N*-ethylmaleimide-sensitive factor attachment protein receptors complex involved in membrane fusion. We induce large conformational changes, carry out interactive docking experiments, probe lipid–protein interactions and are able to sense the mechanical properties of a molecular model. Furthermore, such interactive simulations facilitate exploration of modeling parameters for method improvement. For the purpose of these simulations, we have developed a freely available software library called MDDriver. It uses the IMD protocol from NAMD and facilitates the implementation and application of interactive simulations. With MDDriver it becomes very easy to render any particle-based molecular simulation engine interactive. Here we use its implementation in the Gromacs software as an example.

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Introduction

In 1994, the Sculpt program developed by Surles and collaborators permitted to change the shape of proteins interactively using continual energy minimization.[1] Since this pioneer work, the interactive molecular simulations field has been growing quietly but continuously. Initial interactive experiments using molecular mechanics techniques gave quickly rise to "guided" dynamics simulations[2,3] or steered molecular dynamics (SMD).[4,5] The interest in these methods increased with the enhancement of simulation accuracy. Thanks to technological and methodological progress, exciting new possibilities for dynamic structural exploration of very large and complex models of biological systems came within reach. In the emerging interactive molecular dynamics (IMD) approach, steering forces are applied to the molecular system in interactive time, with a chosen amplitude, direction and application point. This approach enables the user to explore the simulation system while receiving instant feedback information from real-time visualization or via haptic devices.[6,7]

Additional Supporting Information may be found in the online version of this article.

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Recently, Schulten's group has carried out several convincing studies where IMD simulations were applied to macromolecular structures.[8,9] This effort led to the design of two innovative, efficient software tools greatly facilitating the process of setting up an IMD simulation: NAMD and visual molecular dynamics (VMD).[10,11] Similar projects proposing an interactive display for molecular simulations exist, such as the Java3D interface,[12,13] the Protein Interactive Theatre,[14] or others.[15,16] New autofabrication technology now even makes it possible to create tangible molecular models that can within certain limits be interfaced with molecular simulations.[17]

Recent Hardware Developments Open New Possibilities

Interactive simulations show a fast and promising evolution, owing to the wide spread of new computer hardware devices and increasing accessibility to powerful computational infrastructures. These techniques can be applied to the study of very large molecular systems comprising hundreds of thousands of atoms. Such applications are now in the reach of state-of-the-art desktop computing. This evolution was possible given the strong increase in raw computing power leading to faster and bigger processing units (multiprocessors, multicore architectures).[18] At the same time, important progress was made in simulation algorithms and methods. Currently ongoing technological developments such as graphics processing unit (GPU) or physics processing unit (PPU) computing[19–23] and the spread of parallelizable entertainment devices (PS3, Cell) with specific graphic and processing capabilities open exciting new opportunities for interactive calculations. These approaches could provide even more processing power for highly parallelizable computational problems - for instance by differentiating the parallelization of molecular calculations and graphical display functionalities. Given these developments, the range of computational methods and representations accessible to interactive simulations is bound to grow. It may soon be possible to extend the interactive approach to *ab initio* or quantum mechanics/Molecular mechanics (QM/MM) calculations. Such refined physical modeling methods increase the accuracy of the calculation. Alternatively, the precision of a given model can be improved by a better description of the molecular context and environment. This can be achieved via multiscale simulations,[24] which would largely benefit from interactivity for several reasons. First, the systems under study are often complex and the corresponding molecular models require fine-tuning of the interactions, which is difficult to achieve by noninteractive trial and error. Second, by using multiple scales it is possible to focus on the essential degrees of freedom of a complex biological system. This makes the user's interaction with the system more efficient. However, the raw increase in computer speed alone is not sufficient to grant a successful future evolution of these approaches. In addition, it is necessary to develop adapted software solutions, which are generally more efficient.[8]

Overview

In this article, we will focus on IMD as a textbook example for interactive simulation methods. We will point out how this approach can be used to explore new facets of molecular assemblies and gain an intuitive insight into the underlying physical properties of the model. Interactive simulations can thus be very useful in exploring and generating hypotheses about the structural and mechanical aspects of biomolecular interactions. This will be illustrated using a range of interesting biological applications including interactive docking. Possible applications further include method improvement and new developments. In this context, interactive simulations can facilitate exploration and adjustment of modeling parameters. Currently, remaining unsolved intrinsic problems with IMD will be discussed. In all related studies published up to date, all-atom molecular dynamics simulations were employed. In this study, we extend the application of the IMD approach to simplified modeling methods such as coarse-graining.[18,25–28] These methods allow for larger integration timesteps and thus span a much longer lapse of simulated time during the same real duration compared to classical simulations at atomic detail. Hereafter, we show that these techniques appear to be more suitable for interactive experiments. They represent a well-balanced compromise between an important gain in computational speed versus a moderate loss in modeling accuracy compared to higher resolution all-atom simulations. At the same time, coarse-grained IMD enables exciting applications thus helping to solve important biological and biochemical problems. This low resolution approach is particularly useful for initial exploration and hypothesis development with respect to rare molecular interaction events.

In the near future, the development of interactive simulations will surely allow to further increase the appeal of such an approach. One might imagine several lines of novel applications such as instantaneous data analysis, on-the-fly free energy computations or intuitive docking facilities guided by human interaction. Maybe it will even become possible to guide simulated chemical reactions interactively. In this perspective, we have tried to evaluate which applications are currently feasible. For this purpose, we have explored combinations of all-atom or coarse-grained modeling methods implemented in the Gromacs software package within IMD simulations driven via the VMD tool.[29] We have tested this approach on a large selection of complex molecular assemblies using a library prototype called MDDriver. The MDDriver software library was developed as a generic platform that can be used with any particle-based simulation code for rapid prototyping. It's purpose is to facilitate the implementation and application of interactive molecular simulations. Only minimal changes to existing software are required. Using MDDriver, it becomes easy to combine both a virtual reality-enabled visualization interface and a user's preferred calculation code, using a standardized and optimized treatment for the particle coordinate exchange. We hope that this freely available tool will spawn further development of the method and lead to a broad spread of interactive simulation methods among the computational chemistry and molecular modeling communities.

Materials and Methods

Technical Implementation

During the design of the MDDriver library, initial tests have been carried out with the VMD/NAMD ensemble. We characterized the data exchange occurring between both modules. The

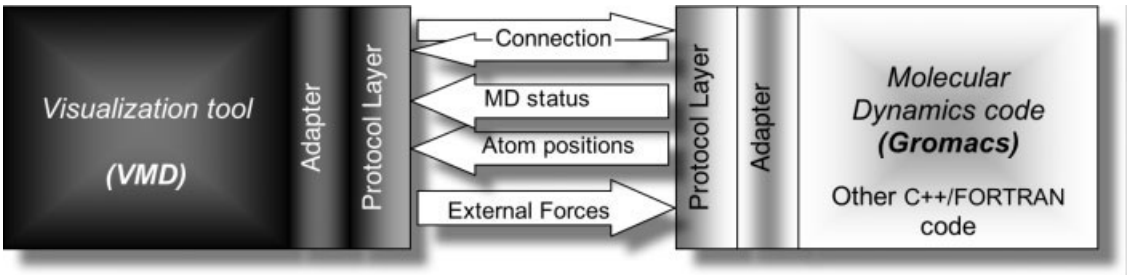


Figure 1. Data flow between a molecular visualization tool such as VMD and a generic simulation code, here molecular dynamics via Gromacs. The central arrows represent the main data flow, connected to a generic protocol layer. The subsequent adapter layer is in charge of adjusting the exchanged data for compatibility with a legacy MD application (or inversely to adapt the MD application data to the protocol layer).

communication mechanism is based on TCP/IP sockets and uses a specific "IMD protocol". Further analysis revealed that this approach is simple, robust and extensible. We have thus based the central communications within the MDDriver library on this previously developed protocol, preserving a complete compatibility with the NAMD-VMD tools. Figure 1 and Supporting Information Figure 1 illustrate how MDDriver couples a visualization and a calculation module. The IMD protocol was extracted from NAMD and encapsulated in an abstraction layer. The user part of this protocol layer consists of only five high-level subroutines shown in Supporting Information Table 1 and can potentially be called from any C/C++ or from FORTRAN code. This layer is mainly configured with the network port identifier and the coupling frequency (or data refreshment rate). As in the original IMD protocol, the MDDriver library includes integrated control functionalities such as pausing or stopping the simulation from the graphical user interface, and dynamically changing the data transfer rate. In addition, we have implemented a log message functionality that displays information on the standard output or writes it to a file using different verbosity levels. This functionality could easily be extended to record the user interaction for further analysis or replay. If the user wants to stop an interactive manipulation and pick it up at a later time, he can either pause the simulation or create a restart file at regular intervals. This functionality could be enhanced in a future version of the MDDriver software library.

An intermediate interface adapts the data from the protocol layer MDDriver library to the corresponding data structure in the application code, here Gromacs (see Supporting Information Fig. 1). This adapter is specific for a given simulation software. It is mainly in charge of transforming the physical values into adequate units, but it could also operate more complex transformations, such as changing data from a given discretization model to another one. The adapter is also in charge of redistributing and collecting the data in a parallel application. For implementing MDDriver in a given software package, the major part of the work consists of locating at which point in the simulation algorithm the data exchanges need to take place. The exchange with the visualization tool includes sending atom positions, receiving forces, and managing control events for the simulation.

Table 1. Characteristics of the Simulation Systems Used as Testbed for Validating the *MDDriver* Gromacs Implementation.

Nb.	Number of particles	PDB ID Code[a]	Resolution	Description
S1	1900	1S4Q	Coarse-grain	Guanylate kinase in water
S2	2741	1SRN	All-atom	S-peptide (residues 1 to 19) from semisynthetic ribonuclease in water
S3	6184	–[b]	Coarse-grain	Synaptobrevin-TMD in a DPPC lipid bilayer
S4	18,098	1S4Q	All-atom	Guanylate kinase in water
S5	29,965	1QD5	All-atom	OmplA monomer in an octane slab
S6	33,028	1SFC[c]	Coarse-grain	Full SNARE complex in a hydrated double DPPC lipid bilayer
S7	44,196	1QD5	All-atom	OmplA monomer in a POPC lipid bilayer
S8	55,607	1QD6	All-atom	OmplA dimer in a POPC lipid bilayer
S9	65,118	1I78	All-atom	OmpT enzyme in complex with ARRA peptide in a DMPC lipid bilayer
S10	72,466	1SFC	All-atom	SNARE core complex in water
S11	169,331	1N7D	All-atom	Low density lipoprotein receptor in water at pH 5
S12	304,564	1SFC[c]	All-atom	Full SNARE complex in a hydrated double lipid bilayer (mixed POPC/POPS)

[a]Protein Databank ID code of the starting structure used for building the model.
[b]The model was extracted from the model of Ref. 39.
[c]The full model was taken from the work described in Ref. 39.

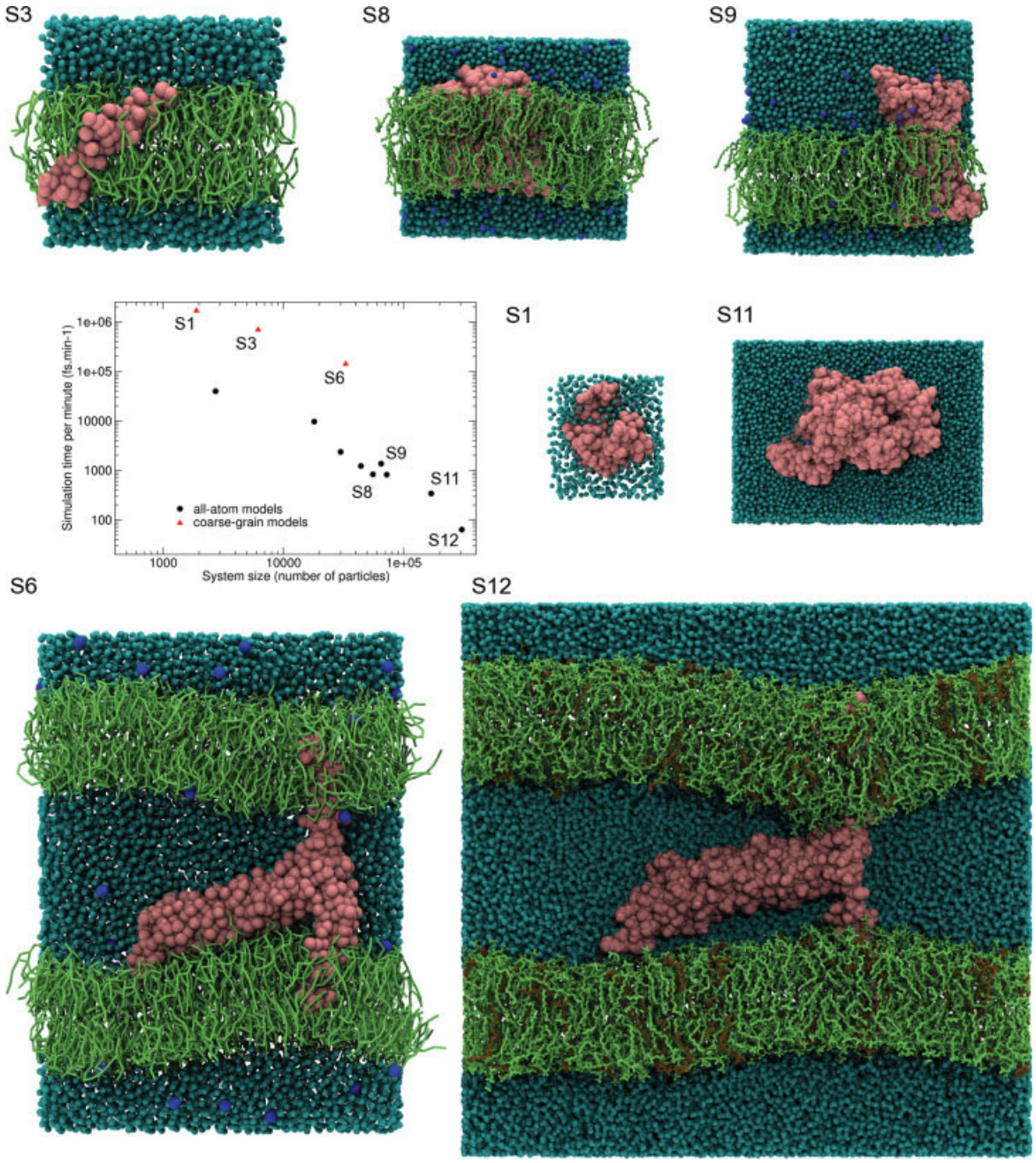


Figure 2. Illustration of some of the 12 test systems used for the validation of the MDDriver Gromacs implementation. The systems are described in detail in Table 1. The central graph indicates the distribution of system size versus the number of simulation steps per minute, a measure that describes how suitable the simulation speed is for interactive experiments. All-atom models are shown as black squares, coarse-grain models as red triangles. Selected snapshots are shown surrounding the central panel with proteins in pink, water in light blue, lipids in green, and other particles such as ions in blue. System 6 is a coarse-grained representation of system 12.

Such an adapter has been implemented for the Gromacs code and allows for keeping the changes to the initial MD code at a minimum. Only about 25 lines of the original Gromacs code needed to be changed (the Gromacs adapter which must redistribute and collect data, contains less than 500 lines). A detailed flow diagram illustrating the interactions between Gromacs and the MDDriver adapter layer is provided as Supporting Information Figure 2. The implementation of these changes in the

source code has been achieved in collaboration with the main Gromacs developers. The adapter implements control functionalities similar to VMD/NAMD. Three ways are provided to control a Gromacs simulation. The pause mode freezes the simulation until a new pause event occurs. The disconnection mode stops the interaction with the MD code while the simulation goes on—a follow-up connection is allowed. The kill mode properly stops the Gromacs code (output files are stored) and a subsequent simulation can be run with the adequate Gromacs input files. We have validated this approach in the light of prospective IMD simulations deployed on a large computing infrastructure (DEISA). These tests allowed us to point out some critical features related to performance loss due to parallelization and data exchange (the atom position transfer rate must be tuned), as well as to define the limits in the graphical rendering of a moving system in a supercomputing context. More technical details can be found in the MDDriver source code itself, in its accompanying documentation and tutorials.

The part of the MDDriver library presented in this article was developed for carrying out Gromacs IMD simulations using either all-atom or alternatively coarse-grained modeling methods. The final application is composed of a server part interfacing the simulation engine with the client part, the VMD visualization tool. The modularity of the MDDriver library permitted us to adapt the implementation template designed for the Gromacs application to several other simulation codes exchanging coordinates and forces. It is thus easy to extend any such simulation engine with an interactive interface. The use of the VMD program allowed us to benefit from an efficient interactive visualization tool with an evolved interaction interface. Furthermore, VMD includes bindings to the virtual reality peripheral network (VRPN) library[30] enabling an easy dialog with virtual reality (VR) devices. The haptic rendering of feedback forces in our coarse-grained IMD simulations relies on the same metaphors as initially implemented by the VMD developers.[11]

Validation on an Extended Benchmark

We have tested our MDDriver-based IMD approach using the implementation in the Gromacs code on a diverse selection of biological systems (see Fig. 2 and Table 1), ranging from very small to very large macromolecular assemblies. The corresponding models were constructed from the experimental structures given in Table 1; additional details are provided in the Supporting Information. The benchmark includes a small peptide in a water box, several larger systems composed of membrane protein complexes inserted into fully hydrated lipid bilayers, and a double-bilayer model of the membrane-inserted soluble *N*-ethylmaleimide-sensitive factor attachment protein receptors (SNARE) complex with two mixed palmitoyloleoylphosphatidylcholine (POPC)/palmitoyloleoylphosphatidylserine (POPS) lipid membranes. The sizes for the simulated all-atom model systems range from 2741 to 304,564 atoms. For the coarse-grained systems, sizes range from 1900 to 33,028 particles (for further details see Table 1). In this report, we will focus specifically on three representative systems differing in their extent (from 1900 to more than 300,000 particles) and in the variety of applications that the IMD approach offers.

Large Conformational Changes

The guanylate kinase (GK) enzyme catalyzes the transfer of a phosphate group from an adenosine tri-phosphate (ATP) to a guanosine 5′mono-phosphate (GMP) nucleotide. This system has been extensively studied experimentally.[32–35] The protein shows an interesting structural behavior bearing two main conformations. An open form is observed before the docking of the GMP and ATP substrates or after the release of the GDP and ADP products. A closed form is essential for the enzymatic reaction. The mechanism of GK's closure and opening motions is still not clearly understood.[36] An interactive approach using IMD simulations does provide an innovative and dynamic structural exploration method shining light on these major conformational changes using computational models.

Interactive Docking

Previous molecular modeling studies (unpublished observation),[37,38] have thoroughly investigated the catalytic site of the OmpT membrane protein. This enzyme induces the hydrolysis of di-basic amino acid motifs such as in the ARRA peptide or in a potential autoproteolytic site located on one of its loops. Here we adopt the IMD method to examine the interactive docking of either a substrate molecule or of the L4 loop depicted as autoproteolytic site. Such an investigation does provide important complementary data to previous classical molecular dynamics (MD) simulations.

Protein Mechanical Properties and Lipid–Protein Interactions

The SNARE complex is a proteic assembly involved in the ubiquitous membrane fusion mechanism. This system seems particularly relevant for the study of protein mechanical properties and strong lipid–protein interactions.[31,39] Prior to fusion, the four-helical bundle constituting the SNARE complex is anchored to two opposing membranes that are to be fused. This is achieved via the insertion of two transmembrane domains (TMDs) belonging to two distinct helices of the bundle (syntaxin and synaptobrevin). A model of this state presents an interesting structural entity exhibiting crucial patterns specific to protein–protein and lipid–protein interactions. Using a force feedback device in IMD simulations of this system, we experienced a significantly enhanced intuitive perception of the mechanical behavior of our model of the SNARE complex via haptics.

Hardware and Simulation Protocols

All IMD simulations were performed on a 3GHz dual quad-core MacPro Apple computer using Gromacs (www.gromacs.org)[40] and either all-atom or coarse-grain models.[25–27] Additional performance benchmarks were carried out on the machine zahir, an IBM Regatta Power 4 computer at the IDRIS supercomputing centre in France. The molecular scene display was rendered using an NVIDIA Quadro FX 5600 graphics card. Stereo rendering was achieved via a Crystal Eyes stereovision device. The user interaction occurred via a Phantom Omni haptic device by Sensable Technologies.

Where available, the same simulation protocols as described in the literature were used for our interactive experi-

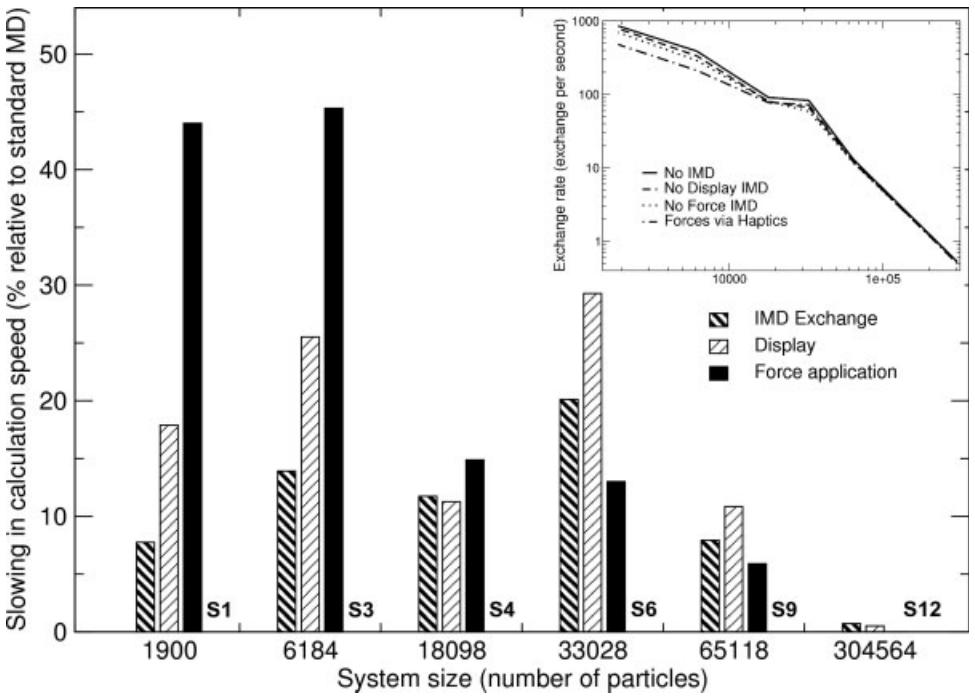


Figure 3. Inset: Benchmark results showing exchange rate in number of coordinate sets per second relative to system size (number of particles). Main plot: The slowdown in the exchange rate compared to classical noninteractive MD is characterized as percentage. Detailed results separated into effects induced by the connection to the MDDriver library (IMD Exchange), the execution of a visualization tool (Display) or the impact of force application. The effect of the user interaction on calculation speed decreases with increasing system size.

ments.[31,37,39,41,42] For unpublished simulations, such as those of the GK enzyme, similar MD protocols were used, based on commonly established standards. The most critical simulation parameters that can directly impact on the simulation performance are the simulation timestep, the neighbor list update frequency and the treatment of long-range electrostatics. The simulation timestep was 2 fs for all-atom and 40 fs for coarse-grain simulations. The neighbor list update frequency was set to every 10 simulation steps. In the coarse-grained simulations, electrostatic interactions were treated with the standard Coulombic potential with a dielectric constant of 20 and a 12 Å cutoff. Shift functions were applied so that the energies and forces vanish at the cutoff distance. For all-atom simulations, a particle mesh Ewald (PME) algorithm was used with a direct space cutoff between 10 and 18 Å. In general, the interactive simulations were run with the same parameters as previous noninteractive ones, without any tuning to enhance interactivity. Generation of the solvent or membrane models and construction of the coarse grain models is described in Supporting Information and Methods.

Results

Performance

We will briefly comment on the performance obtained with the MDDriver library implementation of the Gromacs code in the context of a desktop use. The data transfer rate for coordinates, simulation status and user-applied forces between calculation and visualization modules essentially depends on the size of the simulated system (see Fig. 3, inset). The time spent on handling user forces particularly affects IMD performance for small systems. It depends on the ratio of selected to total particles, which affects the amount of exchanged data. The display rendering step represents the major limitation in the case of larger systems, despite the fact that all measurements were carried out with the lightest representation of the scene appropriate for a given system. In most interactive simulations, we use a cartoon representation for displaying the global structure of the protein in combination with a space-filling display for active sites, binding pockets or substrate residues. With our desktop installation we have noticed two major inconveniences related to latency issues affecting user interaction. One issue originates from the pick and unpick events triggered by the haptic device when the user selects an atom. Another latency issue occurs even in the absence of interaction with the molecular scene. It is a direct effect of the MD calculation protocol, and corresponds to a slow-down every time the neighbor list used for the calculation of the pairwise interactions is recomputed by Gromacs. When the frequency of neighbor list calculation is set to every 10 simulation steps, a commonly used value, one observes a substantial decrease in the IMD exchange rate at precisely this interval (See Supporting Information Fig. 3). Obviously, these latencies in the

interactive molecular scene display do not exist in a classical postsimulation trajectory rendering. In an IMD context, they can have severe effects on the simulation, temporarily freezing the interactive session and perturbing the haptic rendering or more drastically threaten simulation stability.

Several test calculations were also deployed in a supercomputing context using a large computing infrastructure. Fundamentally, the Gromacs IMD simulations using MDDriver showed similar performance characteristics as observed in the context of a desktop computing use. As expected, it was possible to increase the size of the largest system that can still be treated in interactive time. This confirms the robustness of the MDDriver library when coupled to parallelized applications in a supercomputing environment. Generally, one can distinguish three situations (see Fig. 3). For very large systems (exceeding 60,000 particles), the scalability of the code performing the molecular simulation is the dominant performance bottleneck. Between 10,000 and 60,000 particles, network exchange, graphical rendering or user interaction slow down the interactive simulation equally. Below 10,000 particles, the interaction component becomes the biggest limitation, followed by the display component.

Applied Force via Springs

Interaction with the simulation was implemented as in the VMD tool,[11] allowing the user to impose forces on particles and experience a tactile feedback. The haptic device is used to control the direction of the forces applied to selected atoms and to adjust the amplitude of the force. This interaction method contains two stages. The first stage comprises the selection of a single atom or a set of atoms using a 3D tool attached to a haptic device and its buttons. In a second stage, the model described in[9] is used to compute the forces applied to the selected atoms. The main idea of this paradigm is to link the selected atoms and the 3D haptic tool with a spring. Instead of providing direct haptic rendering of forces as computed in the simulation, the force feedback only depends on the spring length, which in turn is influenced by the way the simulation reacts to the applied force. A strong point of this approach is that a low physical simulation framerate (5–10 Hz) does not cause instabilities and does not affect the quality of the haptic feedback. The reason is that the spring length is computed in haptic time, at 300 to 1000 Hz. A new aspect with respect to the haptic feedback concerns the user's perception of forces in a coarse-grain simulation.

Applied Force Parameters

Parameters related to the applied force were previously discussed by the VMD-NAMD developers. The feedback scale factor $\boldsymbol{S}_{\mathbf{fb}}$ and the applied force scale factor $\boldsymbol{S}_{\mathbf{f}}$ can be changed in interactive time. These parameters are the most affected ones upon lowering the resolution of the molecular representation from an all-atom to a coarse-grained level. The origin of this effect lies in the concomitant modification of the simulation timestep and of the potential energy surfaces.[26–28] The user can adjust the $\boldsymbol{S}_{\mathbf{fb}}$ and $\boldsymbol{S}_{\mathbf{f}}$ parameters. This is often required during an ongoing IMD simulation to adapt to the nature of the system under study. Tuning $\boldsymbol{S}_{\mathbf{f}}$ is a necessary but quite delicate task, ideally resulting in improved force efficiency. Yet in the case of an overestimation, this adjustment may induce strong structural distortions and threaten simulation stability. This point will be discussed in more detail below. Furthermore, proper force scaling is tied to a precise and judicious atom selection. This selection defines the application point of the force and is crucial to prevent irreversible or damageable losses of structural integrity. By performing low-resolution coarse-grain simulations rather than more detailed all-atom ones, while choosing well adapted $\boldsymbol{S}_{\mathbf{f}}$ and $\boldsymbol{S}_{\mathbf{fb}}$ parameters, an improved user perception of the haptic feedback is achieved.

Scientific Applications

We carried out a range of interactive molecular simulations using the MDDriver library to establish a dialogue between the Gromacs MD code and the VMD visualization tool. This approach allowed us to explore a large scope of biological and molecular interactions in atomistic and low-resolution models in an intuitive way. The biochemical events that have been examined interactively deal with a broad range of scenarios, including the study of protein–lipid as well as lipid–lipid interactions, enzyme-substrate recognition, macromolecular assembly and disassembly (see Fig. 4). Among all these applications, we will point out a few key findings, illustrating how the IMD approach can lead to the discovery of new and important structural and mechanical properties and thus generates new hypotheses for further validation. We will present an overview of potential applications to the field of structural biology and biochemistry. The interactive simulation approach provides access to a new dimension of complementary intuitive information about computer models of molecular assemblies. We will particularly insist on new features emerging from coarse-grain IMD simulations.

Exploring the OmpT Enzyme's Catalytic Site and Its Substrate Docking

One possible approach for characterizing the key residues for substrate docking in the OmpT enzyme is to observe the relaxation of its active site after a perturbation by the user. This experiment was carried out on the membrane-inserted enzyme using an all-atom model comprising 65,118 atoms. After a short (in user time) force application to guide the ARRA substrate inside the catalytic site of OmpT, the positively charged substrate is spontaneously recognized by surface residues that are part of the active site (D83, D85, and D97). No such recognition is observed deeper inside the enzyme, where the user has to precisely drive basic amino acids to create stable interactions with residues such as E27, D208, and D210. Substrate recognition and long-range interactions do not seem to involve the two basic amino acid residues on loop L4 which could play an autoproteolytic role. No specific conformational change of this loop is observed during docking of ARRA with a variety of user chosen interactive approaches. An alternative experiment is to start from a configuration where the substrate is already docked to the enzyme. The user then perturbs this structure by imposing small displacements. In this case, the user can observe a qualitative hierarchy among the contacts that are created between the

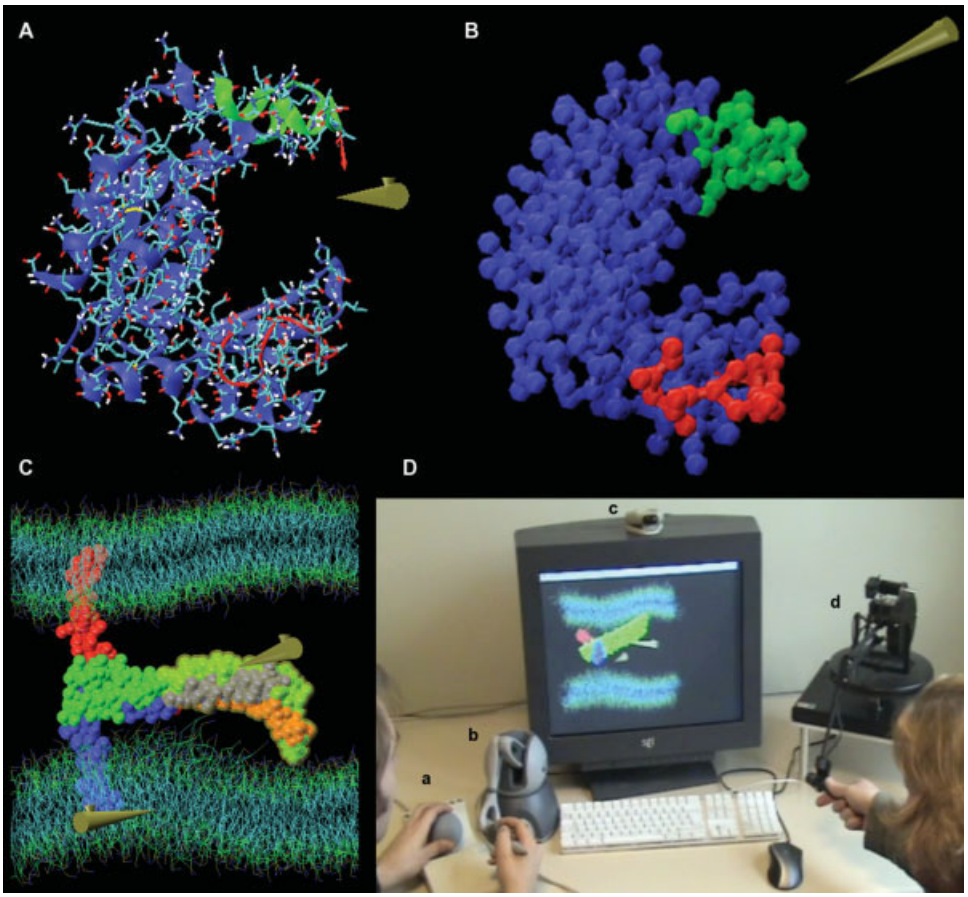


Figure 4. Scientific applications of IMD simulations to structural biology. The top row illustrates how one can induce large conformational changes by exploring the opening and closure mechanism of the U-shaped GK enzyme. IMD simulations were carried out using both an all-atom (Panel A) and a coarse-grain (Panel B) representation of GK. The enzyme is shown in blue with particularly interesting regions highlighted in green and in red color. In Panel A, a red arrow represents applied forces. The bottom row shows how MDDriver was used to probe protein mechanical properties and explore lipid–protein interactions in the SNARE system. Panel D is an example for *in situ* collaborative work where two scientists simultaneously interact with a given simulation. This Panel also provides an overview of the experimental setup used for IMD simulations with several VR devices such as a spaceball (a), a portable haptic device (b), a stereoview rendering equipment (c) and a second haptic stylus (d).

substrate and the different catalytic residues. This hierarchy can be further highlighted by an instant display of hydrogen bond formation and disruption. Strongly pulling the substrate out of the catalytic site leads to nonreversible changes. Concerning *apo* OmpT, we investigate several possibilities for the approach of the autoproteolytic loop L4 towards the catalytic site. It is possible to induce several conformations where the loop resides inside OmpT's active site after crossing a potential barrier. These experiments lead to a variety of new stable states. Instant visual feedback of favorable hydrogen bonds and system energies further helps the user to assess the most favorable hypothesis for such a conformational change. Steering the sidechain of arginine R218 close to the E27-D208-D210 triad seems sufficient to drive the whole loop into the site. This is not the case when a force is applied on lysine K217 towards the D83, D85, and D97 residues, or when a constraint is applied on the full loop backbone. Conducting such tests on a molecular system provides instant feedback. Based on the tactile haptic feedback, the user is intuitively able to assess whether the effect of the move that has been imposed is likely to lead to reversible or irreversible changes.

Inducing Closure and Opening of the Guanylate Kinase Enzyme

The closure mechanism of the U-shaped GK molecule consists of bringing its GMP and ATP binding sites closer together. It is an essential mechanism for the enzymatic reaction and has been investigated via IMD simulations based on both all-atom and coarse-grain methods. The all-atom model comprises 18,098 atoms and the coarse-grain representation consists of 1900

beads. Prospective experiments were carried out on the enzyme's *apo* form using coarse-grain simulations. This approach permitted an efficient and broad exploration of diverse ways to close the enzyme. As a guideline, we tried to reach a closed conformation similar to available homologous structures in the protein data bank. The crucial role of two structural elements in the initiation of GK's closure has thus been identified. Both the loop located between the third and fourth β-sheets (^{56}TRAPRPGEVDGVDY^{69} for *M. tuberculosis* GK) and the fourth α-helix play an essential role in this process. This has been confirmed in a second phase using more accurate all-atom IMD simulations. Hence, we gained an intuitive understanding of this model hinting at an early intermediate state, which seems to provide an impulse for the closure mechanism. These findings allow us to suggest an interesting new mechanistic hypothesis. GMP docking could actually initiate the displacement of the loop. The approach of the negatively charged GMP could drive the loop via long-range electrostatic interactions with two arginine residues located on this fragment. R57 and R60 link the substrate phosphate group to the enzyme and bridge it to the ATP's γ-phosphate at a later stage. This move would favor the global reorientation of the GMP-binding domain. It leads to the creation of new interactions between α-helix four and β-sheet five. Subsequently, specific contacts between the GMP-binding and LID domains are formed, thus achieving a global closure of the enzyme. In summary, a variety of induced conformational changes related to the GK enzyme, some reversible and some irreversible, have been efficiently explored via coarse-grain IMD simulations. The underlying conformational transitions were then refined at a higher atomic resolution.

Exploring the Membrane Anchoring of the SNARE Complex

Simulations of the SNARE complex were carried out with a reduced low-resolution model and with a refined and more intricate all-atom model. The coarse-grain approach comprises 72,466 particles, whereas the detailed representation requires modeling of 304,564 atoms. These simulations have provided twofold feedback. On the one hand, we gained theoretical data on the coarse-grain method that is still under development. Such feedback helps to continuously improve the method. On the other hand, biologically and functionally relevant structural information could be obtained for this system. The transmembrane insertion of the synaptobrevin helix into a lipid bilayer has been extensively studied via interactive simulations on a reduced SNARE system. This allowed us to explore the role of two consecutive tryptophan residues of the TMD. These tryptophans are considered key residues for helix anchoring.[39] Indeed, these residues adopt a firmly anchored stable position in the membrane and resist to a broad variety of perturbation-relaxation trials. Then, we essentially focused on the reorientation and repositioning of specific residues inside the membrane or at its interface with the aqueous solution. After a perturbation by the user it is possible to observe the relaxation of the simulated system and its return to equilibrium. A particularly informative experiment is to pull lipid molecules out of the membrane. Previously, noninteractive simulations were used to quantify the lipid adhesion forces,[44] which require complex postsimulation analysis. The haptic feedback provides intuitive clues about force-field parameterization and the balance of the various forces in the system, such as those between the lipids themselves and between the protein and the lipids. Comparatively, it is rather easy to extract a lipid from the membrane, whereas it is exceedingly difficult to detach the protein TMD.

Probing Mechanical Properties of the SNARE Assembly

Encouraged by the results on membrane insertion of SNARE transmembrane helices, we then examined the larger, extended model of the SNARE system. A pull on the N-terminal part of the four-helix bundle initiates a membrane perturbation. This hints at the possibility of a mechanical functional role of the bundle. It could lead to a perturbation of the bilayer and help catalyze the fusion event. At the same time, one also observes an important stretching of the global protein bundle. This indicates both a structural and a mechanical response of the whole helix bundle to an imposed force. Another observation is that the strength of the insertion of the syntaxin and synaptobrevin transmembrane helices into the two opposing lipid bilayers differs. The IMD approach allowed us to characterize a higher stability for the synaptobrevin transmembrane helix immersed in the vesicular membrane. The syntaxin protein is inserted more weakly into the target membrane. This finding might be a partial explanation as to why the syntaxin membrane anchoring is reinforced by the palmitoyl chains of synaptosome-associated protein of 25 kD (SNAP-25) in a biological context. Finally, the interactive simulations can be used to examine the structural integrity of the SNARE protein assembly. An interesting experiment consists of testing the self-association of the four SNARE helices driven by their hydrophobic complementarity. This can be done by first unzipping the complex, starting from the C-terminal part of the helix bundle, and then following the system while it returns to equilibrium. The synaptobrevin helix appears to be the weakest spot in the bundle, dissociating more readily than the other three helices.

Discussion

The following discussion exposes some of the advances that can be achieved via the IMD method. It is based on the case studies described in this report and on the observation of common features of interactive simulations, new perspectives and possible future enhancements. In particular, we focus on biological applications to the SNARE protein complex, and to both the GK and OmpT enzymes.

Interactive Simulations Provide an Intuitive and Enhanced Perception of Biochemical Systems

In an IMD simulation, the user guides the evolution of a given molecular system to overcome the short timescale of classical MD simulations. In this way, the user can explore rare events by enhancing their probability through his deliberate interactions. Particular benefits from the interactive approach are as follows. Firstly, the specification of a complex set of time varying conditions that drive the simulation is made easy. Second, the contin-

uous perceptual feedback produced during the course of the simulation is very valuable for the user. One can thus compel a nonspontaneous event to probe the related mechanical properties and structural deformations of a model of a given molecular system. Two potential issues concern the exhaustiveness and the reproducibility of the exploration. The approach is not by itself exhaustive; the pertinence of the exploration depends on the user's expertise to determine which explorations make sense. With MDDriver it is however possible to script some simpler deformations to achieve an automated procedure somewhat comparable to a grid search. Reproducibility can be quite delicate depending on how severely the system was perturbed. However, by saving the user interactions to a file for later replay one should be able to achieve a certain degree of reproducibility. This also enables the comparison of the directions and magnitudes of the forces in several separate simulations.

Membrane Anchoring, Interactive Docking and Novel Hypotheses on Membrane Fusion

Through this approach the user can easily explore the reversibility of structural events, induce large conformational changes or sense the mechanical properties of a molecular model. We have presented a first series of interactive investigations via a perturbation-relaxation approach. This technique has been applied to a lipid bilayer in the neighborhood of membrane proteins such as OmpT or synaptobrevin. The response of the TMDs of these proteins supplied new understanding about how these molecular models reproduce their insertion and anchoring into membranes. The positions of hydrophobic residues relative to the lipid chains and the membrane deformation after pulling on the synaptobrevin transmembrane helix have thus been efficiently studied. Such studies lead to new hypotheses that may be validated experimentally. Interactive and intuitive docking is another potential application as shown for the ARRA substrate interacting with OmpT. Multiple pathways for substrate approach were tested with this dynamic method and haptic rendering discussed below allowed for an enhanced perception of their feasibility. Interactive experiments were also informative for the study of enzymatic systems and allowed us to investigate the relationship between biological function and conformational fluctuations. Major conformational changes occurring during the closure of the GK enzyme could be investigated. In the case of the OmpT enzyme, we carried out a steered exploration of potentially correlated conformational transitions coupling the fluctuations of the substrate with those of nearby extracellular loops. In the same way, the reported autoproteolytic reaction of OmpT could be studied by driving the loop self-orientation via IMD simulations.

The study of the full-size model of the SNARE complex at coarse-grained resolution has provided valuable information about key features that might be essential for further understanding of the membrane fusion mechanism. Our experiments aimed at reproducing and exploring the mechanical action of the complex on the lipid bilayer. We realized this by pulling on the N-terminal hydrophilic and solvent-exposed part of the protein bundle. Our aim was to characterize the impact of this remote perturbation on the lipid bilayer. We have also observed a difference in the anchoring strength of the syntaxin and synaptobrevin TMDs in the membrane. This observation could possibly arise from an artefact in the coarse-grain parameterization, but it might as well hint at a functionally significant behavior. Higher resolution all-atom and noninteractive MD simulations will be necessary to confirm this feature. These first results highlight interesting perspectives for future investigations, such as the extension of the model system to multiple SNARE complexes.

Collaborative and Quantitative IMD Simulations

Another exciting application domain concerns cooperative IMD simulations. In particular for the SNARE system, this approach might provide further insight into the creation of a membrane pore. In that case, several users would simultaneously interact with the simulation system, each one pulling on one of the complexes. We did carry out first exploratory experiments on such *in situ* collaborative work using a setup with two haptic devices shown in Figure 4D.

In this report, we have mainly focused on intuitive and qualitative data. It has to be pointed out that reliable quantitative information can be reconstructed from IMD simulations as well. This is possible by recording simulation parameters and data such as trajectories, energy terms and imposed forces followed by postsimulation analysis. The user can for instance compute a potential of mean force from such simulations, as has been described for SMD simulations.[44,45]

Haptic Scene Rendering Provides Additional Benefits with Respect to Visual Feedback

The interactive experiments described in this article can be carried out without any specific equipment, just using a mouse and on-screen visual feedback. However, comparing these simulations with and without haptic feedback has shown that haptic devices provide several benefits. A quantitative evaluation of this question goes beyond the scope of this article, but some qualitative observations are discussed.

Generally speaking, immediate feedback from the simulation adds a new dimension of user perception. The instant visual feedback occurring after a user-induced perturbation is one of the most informative ways to use IMD for perceptive learning. Adding haptic feedback to this instant visualization further increases the understanding of a given system. Indeed, the force resistance provided by the haptic device using a mass-spring model, in addition to the visual feedback, allows the user to experience physical effects such as solvent viscosity or the mechanical resistance of a secondary structure element. This is intellectually easy to assimilate, even for structural biology or molecular modeling neophytes. We have successfully used this approach for teaching purposes, startling students with the high resistance forces preventing the extraction of an R-SNARE fragment from the membrane.

To interact with a simulation in progress, it is possible to use a mouse for adding force constraints on atoms or pseudo atoms. This provides 2 degrees of freedom for the interaction. However, using a haptic device is well adapted to this task, especially for selecting and catching moving atoms. Such a device with 3 instead of 2 degrees of freedom is more intuitive and efficient for interacting with a complex three-dimensional object. Further-

more, the immediate haptic feedback when an atom is actually picked significantly improves the user experience and greatly helps to immerse the user in the molecular scene. When using visual feedback only, the user often asks for additional explanations before getting started. With force feedback, this barrier is lifted, as the interactive simulation is comparable to dextrous manipulations such as those carried out in daily life.

It should also be pointed out that the hardware requirements are modest. In our experience, this approach is viable using a small and affordable haptic device, providing 3D positions and handling 3D directional force feedback. Such an entry-level solution designed for a desktop use is targeted at a large user community and is very easy to set up. The availability of convenient drivers that are compatible with software libraries such as VRPN[30] was a major criterium for our choice. We are also working on more advanced feedback, using six-dimensional output forces with additional torsion force feedback for experiencing more complex properties on biomolecules.

Using Interactivity to Enhance the Development of New Simulation Methods

Another, yet unexplored possibility for IMD simulation experiments concerns rapid and intuitive prototyping of novel modeling methods. This was for instance the case with the coarse-graining approach used in this work. The method is still under ongoing development. Several advantages could be gained from interactivity. After changing a forcefield or simulation parameter, one can visualize the evolution of related physico-chemical properties on the fly (lipid–lipid selfassembly, solvation effects). Feedback on the stability of a simulation with respect to a given conformation (integrity of a fold, heavy metal coordination sphere) is immediate. The impact of parameter or algorithm changes on calculation speed (frequency and algorithm of neighbor list calculation) is observed directly. For instance, we observed a slow-down every time the neighbor list used for the calculation of the pairwise interactions is recomputed by Gromacs (Supporting Information Fig. 3). These are just a few examples.

To the best of our knowledge this aspect has not been explored in preceding studies, focusing essentially on "steering" or "guidance" possibilities of interactive dynamics. Interactive simulations can for instance be used in an exploratory context. This implies less accuracy in the control of the induced force while providing intuitive and instantaneous tactile feedback via haptics. Such feedback illustrates how the program is simulating specific components of the forcefield, for instance independent terms of the potential energy. It can be used to improve a known forcefield for a given system or to test newly developed parameters for atypical residues. Using careful post-IMD analysis, one can furthermore obtain quantitative and reliable data enhancing a scientist's knowledge and perception of the underlying simulation method.

Comparative IMD Simulations at Different Resolutions

A comparative approach, where parameters are calibrated at different resolutions on the same reference system, can be used to improve a given simulation method. We have presented examples using both a detailed all-atom and an approximated coarse-grain representation. Such simulations help to point out discrepancies in the underlying modeling methods and highlight strengths and weaknesses. The IMD user can thus make an informed choice between the accuracy of a higher resolution representation and the flexibility of a low-resolution interactive simulation depending on the specific requirements for the study of a given molecular system. For instance, pulling a synaptobrevin transmembrane helix out of the membrane provided qualitative information about the anchoring strength of the protein fragment within the lipid bilayer. Comparison of the propensity of a single lipid molecule or of the TMD to return into the membrane with all-atom simulations indicated an excellent agreement with the coarse-grain model parameterization. This validated the more exhaustive and broader exploration carried out at the CG level. Simulating a system with an IMD based approach at different modeling scales also provides information about the accuracy, the pertinence or the implementation of methods in terms of reliable energies and physical or chemical properties.

Table 2. Simulation Time Explored During an Interactive Experiment of One Minute Duration Using IMD via *MDDriver*.

System	Number of particles	Resolution	Simulation time (ps)
Guanylate kinase	1900	Coarse-grain	17,000
	18,098	All-atom	10
SNARE complex	33,028	Coarse-grain	100
	304,564	All-atom	0.06

Two biological systems are examined using either a high-resolution (all-atom) or low-resolution (coarse-grain) representation.

Scripting Simulations via IMD

Finally, another aspect concerns the programmatic interaction with a molecular modeling algorithm such as a molecular dynamics simulation. The modularity of the MDDriver library permits an easy development of tools that interact with a calculation at each simulation step. This can for instance be used to compute certain parameters from the system coordinates and reinject external forces depending on these parameters into the simulation. With respect to the applications presented here, one could for example use this approach to impose a series of successive opening/closure events to the GK system.

Interactive Simulations at Coarse-Grained Resolution: A Well-Balanced Choice

Using a coarse-grain representation for interactive simulations provides two marked advantages. It permits bridging an expanded timescale by using large integration timesteps (see Table 2 and the central panel of Fig. 2) and leads to a smooth potential energy surface. These features facilitate imposing large conformational changes on a molecular structure. In comparison, at atomic resolution, one may need to apply exceedingly high forces over a short time interval to be able to observe a conformational change in interactive time. This principal inconvenience with high-resolution all-atom simulations is due both to

the rugged shape of the potential energy surface harboring many local minima and to the limited time-scale of these simulations. The high forces imply an elevated risk of loosing structural integrity. Furthermore, some large conformational changes cannot even be reached with all-atom IMD simulations because of their amplitude. Currently, the short integration timestep and the large number of particles that is required to model a complex biological system do in fact limit the application range of the IMD technique to all-atom simulations. To effectively observe large structural changes with all-atom simulations, noninteractive MD simulations are now performed for hundreds of nanoseconds or more. However, corresponding IMD simulations are intrinsically restricted to the time the user can stay in front of a computer. Typically the user's human timescale for interaction ranges from seconds to minutes. Implementing coarse-graining into IMD does significantly help to hedge these critical points and enables a better and quicker exploration of reversible events as the simulation system is more reactive. Furthermore, it enables the study of even larger systems. The two examples provided in Table 2 are particularly compelling. More than five orders of magnitude in the molecular time-scale can be bridged in a one-minute interactive experiment. For each system—the small GK enzyme and the four-helical SNARE bundle—the low resolution simulation spans a time-scale ~1700 times longer. The increase in integration time-step from 2 fs in the all-atom simulation to 40 fs represents a factor of 20. The roughly ten-fold decrease in the number of particles is responsible for the remaining speed-up.

CG IMD Simulations Do Not Require Excessive User Forces

A low-resolution approach significantly alleviates the problem related to high user-applied forces encountered for all-atom IMD simulations. At high atomic resolution, an effective event is produced with comparable speed (a few seconds at the user's timescale) after a force application with a scaling factor of around 30–50. This refers to the "Applied Force Scaling" tab in the "Tool controls" menu of the VMD software and corresponds to the $\boldsymbol{S}_{\mathbf{f}}$ parameter described above. This scaling factor is reduced to 1–5 in an equivalent CG simulation. In an all-atom IMD simulation, such an interactively imposed move has to be applied during more than 10 min (at a user's timescale) with a moderate force to preserve important secondary structure elements. This was for example the case with respect to the short beta sheets of the OmpT enzyme during the attempt to close its L4 loop.

Overall, a coarse-grained approach represents a well-balanced compromise. Some of the precision of the model is sacrificed to be able to manipulate the simulation at time-scales that are pertinent with respect to biological function. Hence, interactive CG simulations contribute to a better understanding of models of biological systems but also of the underlying modeling methods used for simulating a molecular system as already pointed out above.

Interactive Simulations, a First Step Toward Human Computation in Structural Biology?

In a particularly nice and enjoyable way, the "Fold It!" project (http://fold.it) has recently brought interactive simulations to the fore. "Fold It!" is a 3D-puzzle desktop game, in which the user's task is to fold proteins interactively and without any knowledge prerequisites. The puzzle is essentially based on an interactive-intuitive learning process. The protein folding work carried out by the "Fold It!" users can then be gathered and analyzed. This form of human computation[46] enables large-scale and intelligence-driven sampling of folded protein conformations which is yet beyond the reach of brute force computational approaches.

Conclusion

The MDDriver library introduced in this article offers a simple, modular and generic solution for combining any coordinates-based calculation code with various visualization programs. The interactive simulation method investigated in this work is a powerful tool for the exploration of biomolecular structures in large biological systems and greatly facilitates the generation of new hypotheses. It is now easily accessible, even via an affordable desktop computing environment.

The IMD simulation improvements presented here allow some biologically relevant questions to be addressed. More generally, if we consider our selection of systems, several types of applications can be suggested. They comprise protein/substrate interactions and the impact of long-range forces on steering and docking of substrates, but also local interactions such as those inside a catalytic pocket. It is possible to investigate stabilization, mechanical properties linked to functional-decisive conformational equilibria and the structural/mechanical behavior of domains in large macromolecular assemblies. Many other exciting applications are expected.

Simulations implementing a coarse-grain representation offer new perspectives and enable even more challenging interactive experiments. The low resolution allows one to increase the accessible time-scale for following the evolution of a given system. It becomes quicker and easier to manipulate structures during an IMD simulation. The relaxation process after a perturbation by the user provides instant information about the underlying calculation methods and force fields. An interesting evolution would be to design multiscale or multiphysics IMD methods offering the best compromise between accuracy and rapidity of the simulations.

In the context of methodological developments, IMD does also appear as a powerful tool to improve new methods under development or to fine-tune critical simulation parameters. We encourage experimenting with the MDDriver framework, which will soon be available at http://mddriver.sourceforge.net with technical documentation, movies and tutorials.

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